DAY 45

Model Versioning & Experiment Tracking

Beginner-friendly guide • Practical Python • Tasks and solutions

Learning objective

Learn to track experiments, compare model candidates fairly, record settings and metrics, and save a selected model with useful metadata.

Roman Urdu: Har experiment ka record rakho taake baad mein models ko compare, reproduce, aur zaroorat par rollback kar sako.

Repository: https://github.com/DeveloperDesigner-star/Datasciencepdfs

1. Model Versioning Kya Hai?

Model versioning means giving models or model releases clear identities and preserving information about how each was produced. A useful version record may include the model file, dataset identifier, code version, parameters, metrics, and creation time.

Roman Urdu: Model versioning ka matlab har model ko pehchan dena aur us ka record rakhna. Agar naya model kharab perform kare, to purani approved version par wapas ja sakte hain.

• Reproducibility: understand how a model was created.

• Comparison: evaluate candidate versions consistently.

• Rollback: return to a previously approved version.

• Auditability: identify which data and settings produced a result.

2. Experiment Tracking Kya Hai?

An experiment is one training attempt with a particular configuration. Experiment tracking records what was tried and what happened.

Record	Meaning	Example
Parameter	Training setting	n_estimators = 100
Metric	Performance measure	MAE = 21.7
Dataset info	Data used	dataset v3, split seed 42
Artifact	Saved output	best_model.joblib
Run metadata	Run details	ID, timestamp, notes

3. Parameters vs Metrics

Parameters are settings used to train a model, such as number of trees or maximum tree depth. **Metrics** are calculated results, such as MAE, RMSE, precision, recall, or accuracy.

Roman Urdu: Parameter model ki setting hoti hai. Metric batata hai model ki performance kaisi hai. n_estimators parameter hai, MAE metric hai.

For regression, Mean Absolute Error (MAE) is the average absolute difference between actual values and predictions. Lower MAE is usually better when the evaluation setup is the same.

4. Fair Model Comparison

If two candidates are evaluated on different splits, one may appear better simply because its examples were easier. Keep the split consistent or use a carefully designed cross-validation procedure.

Good practice: fit on training data, choose/tune using validation data, and reserve a separate test set for final evaluation. The practice script is a beginner demonstration using one fixed holdout split, not a production selection protocol.

5. Example Experiment Log

Experiment	Parameter	MAE	Meaning
RF-50	50 trees	24.1	Candidate A
RF-100	100 trees	21.7	Lowest example error
RF-200	200 trees	22.3	Not lowest here

These scores are illustrative only and are not the output of the included script.

6. Experiment Tracking Workflow

• Define the problem and evaluation metric.

• Record dataset identity and split strategy.

• Train a candidate with documented parameters.

• Evaluate it using the same validation setup as other candidates.

• Log the parameters, metrics, dataset/code information, timestamp, and notes.

• Compare candidates and select using validation results.

• Save the approved model artifact and metadata.

• Evaluate the final choice on the reserved test set.

7. Run the Python Practice

The accompanying .py file loads scikit-learn's diabetes regression dataset, creates one reproducible split, trains Random Forest models with 50, 100, and 200 trees, calculates MAE, writes a CSV log, and saves the lowest-MAE candidate.

```
python -m pip install pandas scikit-learn joblib\npython Day-45-Model-Versioning-and-Experiment-Tracking-Practice.py
```

Expected output pattern (scores may vary slightly by library version):

```
rf-trees-50: MAE = ...\nrf-trees-100: MAE = ...\nrf-trees-200: MAE = ...\n\nExperiments sorted by MAE:\n...\nSelected: rf-t
```

The script creates experiment_log.csv, best_model.joblib, and best_model_metadata.json inside day45_outputs.

8. Code Walkthrough

• load_and_split_data(): loads the data and makes a repeatable split using a fixed random seed.

• The loop: tries three values for n_estimators.

• fit(): trains the model using training data.

• predict() and mean_absolute_error(): generate predictions and calculate the metric.

• records: stores one dictionary per experiment.

• to_csv(): persists the experiment log so it remains after the script ends.

• joblib.dump(): saves the selected trained model. Only load model files from trusted sources.

• JSON metadata: records the chosen experiment and selection rule.

9. Common Mistakes

• Comparing scores from different data splits or different metrics.

• Saving the model but not recording its data/settings context.

• Repeatedly choosing settings using the final test set.

• Assuming the lowest MAE is always best for every business requirement.

• Thinking a filename like final_model_v7 is enough version documentation.

• Assuming a basic CSV file has all features of a dedicated tracking system.

10. Tool to Explore: MLflow

MLflow is a widely used open-source tool for tracking experiments and managing model lifecycle workflows. It can store parameters, metrics, artifacts, and run metadata. Website: https://mlflow.org/. A CSV log is fine for learning the basics; a dedicated platform helps when experiments and teams grow.

11. Practice Tasks with Solutions

Task 1: Candidate A has MAE 18.5 and Candidate B has MAE 15.2 on the same evaluation data. Which has lower error?

Solution: Candidate B because 15.2 is lower than 18.5.

Task 2: Is n_estimators a parameter or a metric?

Solution: A parameter. It controls the number of trees in a Random Forest.

Task 3: Name four useful experiment-log fields.

Solution: Experiment ID, parameters, metric, and dataset/version information. Timestamp and code version are also helpful.

Task 4: Why not repeatedly choose models using the final test set?

Solution: Decisions can become tuned to the test set, making its score less trustworthy. Use validation data for selection.

Task 5: Artifact vs metric?

Solution: An artifact is a saved output such as a model file. A metric is a measured number such as MAE.

12. Quick Revision and Self-Check

• Versioning identifies and manages model versions.

• Tracking records what was tried and its outcome.

• Parameters are settings; metrics are measured results.

• Consistent evaluation makes comparisons more meaningful.

• Save artifacts and metadata, not just scores.

• Use validation data for selection and a separate test set for final evaluation.

Quiz: What problem does versioning solve? Is n_estimators a parameter or metric? Why keep the evaluation setup consistent? What belongs in an experiment log? When should the final test set be used?

13. Glossary

Term	Simple definition
Experiment	One training attempt.
Parameter	A model or training setting.
Metric	A number measuring performance.
Artifact	A saved output such as a model file.
Reproducibility	Ability to repeat a workflow from recorded details.
Rollback	Return to an earlier known version.
Validation set	Data used to compare/tune candidates.
Test set	Reserved data for final evaluation.